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Neurocognitive Responses to Exam Questions: A Pilot Study Using Functional Near-Infrared Spectroscopy (fNIRS) and Brain-Computer Interaction (BCI)

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Experimental and Participatory Research (ccEXPAR)

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1 Introduction: The Cognitive Divide Between Abstract and Concrete Reasoning

Understanding how students process abstract and concrete information is fundamental to cognitive and pedagogical sciences, education, and assessment design. Abstract reasoning involves the ability to think symbolically, generalize principles, and use concepts that are not immediately tied to tangible experiences. It is essential for higher-order cognitive skills such as mathematical reasoning, problem-solving, and scientific inquiry. In contrast, concrete reasoning is grounded in direct, sensory-based experiences, relying on observable, specific, and real-world contexts. For example, identifying the main idea in a short story, navigating a map, or measuring ingredients for a recipe all draw on concrete reasoning skills. While both forms of reasoning are vital, students exhibit varying degrees of proficiency in each, significantly influencing their learning trajectories and academic performance.

Recent education trends worldwide put a growing emphasis on importance of abstract skills [OECD, 2023]. Yet education systems often assume that students naturally transition from concrete to abstract thinking as they progress through formal schooling. However, this assumption is not universally valid. Research suggests that many students struggle with this cognitive shift, particularly in more advanced areas of mathematics, physics, and philosophy, where abstraction is central [Sweller, 2011]. This cognitive gap can lead to disparities in academic achievement, disproportionately affecting students who rely more heavily on concrete reasoning and struggle to make a switch. Skill assessments, which often prioritize abstract thinking, may therefore not accurately capture the cognitive strengths of all learners [Gardner, 2008].

Current educational frameworks continue relying more on theoretical models of cognitive development (e.g., Piaget's & Vygotsky's theories of cognitive development
[Piaget et al., 1952, Vygotsky and Cole, 1978], both originally from the field of psychology). However, these frameworks often lack empirical evidence from other scientific fields, such as cognitive neuroscience, which focuses on brain engagement during learning [Ansari and Coch, 2006].

In educational assessment, categorisation of test items continues relying on theoretical and pedagogical approaches (e.g. Bloom's taxonomy). Interestingly, there still exists a scientific gap with respect to corroboration of such theories with the help ground data. Without empirical testing on how different assignment questions engage the brain, educators and policymakers risk making decisions based on assumptions rather than neurocognitive evidence. This project seeks to address this gap by leveraging Functional Near-Infrared Spectroscopy (fNIRS) to provide ground truth data on how the brain responds to different cognitive demands in assessment settings [Herff et al., 2014].

2 Implications for Education and Assessment

The insights gained from this interdisciplinary study offer the potential to advance innovation in education-related fields and reshape educational assessment by providing empirical evidence on how students engage with different types of questions. Key implications include:

- **Personalized Learning Strategies:** By identifying the cognitive mechanisms underlying difficulties with abstraction, educators can develop targeted interventions tailored to students' reasoning strengths.
- **Fairer Assessment Design:** Traditional testing methods may disadvantage students who excel in concrete reasoning but struggle with abstract concepts. Empirical fNIRS data can inform the development of more balanced assessments.
- **Curriculum Development:** Understanding how abstract and concrete reasoning develop at the neural level can guide curriculum adjustments to facilitate smoother cognitive transitions for students.
- **Policy Recommendations:** Data-driven insights can inform policymakers on how to structure educational systems to support diverse cognitive profiles, ensuring that no group of students is unfairly disadvantaged.

3 Applying fNIRS to Assess Cognitive Processing in Real-Time

Functional Near-Infrared Spectroscopy (fNIRS) is an advanced neuroimaging technology that allows researchers to monitor brain activity in real time by detecting changes in blood oxygenation. Unlike traditional neuroimaging techniques such as functional Magnetic Resonance Imaging (fMRI), fNIRS is portable, cost-effective, and permits participants to engage in more naturalistic problem-solving scenarios. This makes it particularly well-suited for studying cognitive processes in real-world educational environments.

This study will use fNIRS to investigate how different question types (ranging from highly concrete to deeply abstract) activate specific brain regions associated with cognitive processing. Existing fMRI research by Gilead et al. [Gilead et al., 2014] indicates that processing abstract concepts engages regions such as the prefrontal cortex, while concrete reasoning is more closely associated with sensorimotor and visual processing areas. By applying fNIRS, we aim to validate and expand upon these findings with real-time, dynamic data, capturing how neural activation changes during actual assessment scenarios.

Recent work by Yin et al. [Yin et al., 2025] demonstrated the sensitivity of fNIRS to differentiate between metacognitive, affective, and neutral feedback during chatbot-mediated learning. Their results showed increased activation in the frontopolar and middle temporal regions when learners received metacognitive prompts—areas associated with reflection and semantic integration. These findings confirm the suitability of fNIRS to capture subtle neurocognitive dynamics in educational contexts, providing a methodological precedent for our own investigation of reasoning types and question formats.

Additionally, this research integrates Brain-Computer Interaction (BCI) methods to explore how neural signals vary in response to different types of cognitive challenges. By linking neural data with AI-driven natural language processing (NLP), we aim to identify linguistic and structural features that correlate with specific neural activations, further refining our understanding of how question complexity influences cognitive engagement.

Figure 1: fNIRS device

4 Innovative Aspects

This study represents a novel application of fNIRS technology at LISER to explore neurocognitive responses to different types of exam questions. By combining neuroimaging with AI-driven NLP, we will analyze linguistic and cognitive factors that influence brain activation patterns. This interdisciplinary approach positions LISER at the forefront of neuroeducational research and enhances its research capabilities in cognitive and behavioral sciences.

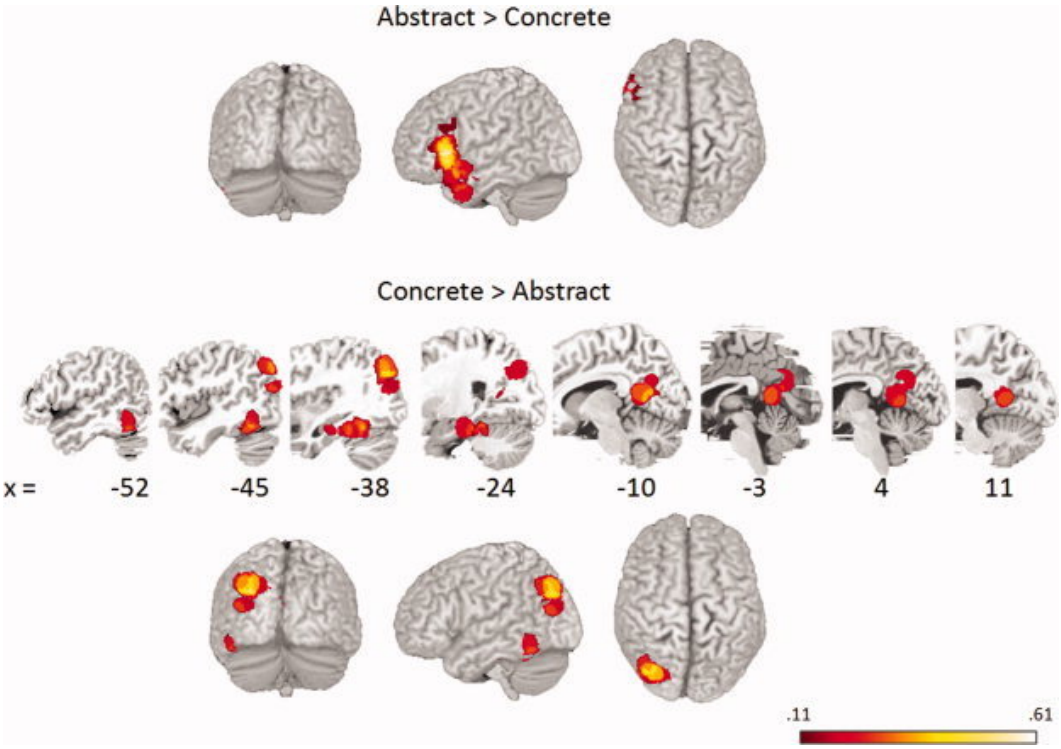

Figure 2: Regions surpassing the maximum activation proportion expected under the null hypothesis, within 10 mm of each voxel for abstract - concrete and concrete - abstract comparisons ($P < 0.05$ FWE corrected, cluster extent).[Gilead et al., 2014]

5 Project Description

We propose a pilot study to measure brain activation while they engage with various exam question types. The study will consist of following stages:

1. **Participant Recruitment** – We aim to recruit with **30 participants**
2. **Neuroimaging via fNIRS** – Real-time monitoring of brain activity during problem-solving tasks.
3. **Comparative Analysis** – Identifying differences in neural activation across different question formats.
4. **AI-Driven NLP Analysis** – Extracting linguistic and cognitive features that correlate with specific neural activations.

5. **Educational Assessment Framework Development** – Translating fNIRS findings into strategies for improving assessments and targeted interventions.

Each participant will undergo a one-hour session, during which they will answer structured sets of questions while wearing the fNIRS device. The questions presented will be real exam questions from previous assessments, for which we utilise publicly-available ground truth data and the socio-economic background of three million students who have previously answered these questions. The collected data will be analyzed to determine which brain regions are activated under different testing conditions.

Research Questions and Hypotheses

This study uses fNIRS to investigate how students engage with abstract versus concrete reasoning tasks during cognitive processing in educational assessments.

The research aims to address the following research questions (RQ) and hypotheses (H):

- **RQ1:** How does the brain respond to abstract versus concrete reasoning tasks in educational assessment contexts?
  - **H1:** Abstract reasoning questions evoke greater brain activation in the prefrontal cortex than concrete reasoning questions.
  - **H2:** Concrete reasoning questions evoke greater activation in brain regions involved in visual and sensory processing than abstract reasoning questions.
- **RQ2:** Which brain regions are differentially activated when participants process abstract compared to concrete questions?
  - **H3:** Participants who perform better on abstract reasoning tasks will show greater activation in executive function areas (e.g., the prefrontal cortex).
- **RQ3:** Can structural features of exam questions (e.g., more complex or abstract language) predict the neural activation patterns observed via fNIRS?
  - **H4:** Exam questions with more complex or abstract language are expected to trigger distinct patterns of brain activation, compared to simpler or more concrete language.

6 Expected Outcomes

- Expertise development in utilizing LISER's new fNIRS device.
- A dataset of neurocognitive responses linked to student performance data.
- AI-driven analysis of neorocognitive response data.
- Contribution to evidence-based policies for students struggling with abstraction and concrete reasoning.
- Foundation for future large-scale research projects and external funding applications.

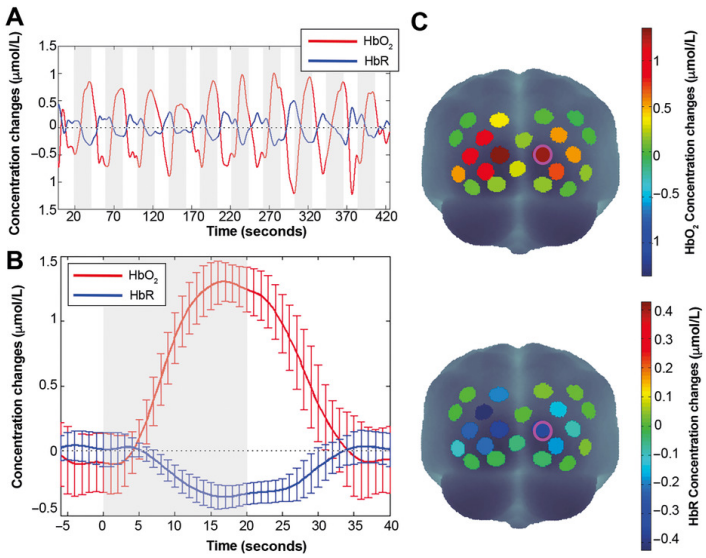

Figure 3: Example of HbO2 (red) and HbR (blue) signals from a representative channel (circled in magenta in panel C) of a single subject over the visual cortex using the fNIRS. The gray areas refer to the stimulation period. Panel B shows the block-averaged hemodynamic response (mean ± SD) computed by averaging the HbO2 and HbR signals presented in A across the 10 task blocks. It is characterized by simultaneous HbO2 increase and HbR decrease. Panel C presents the distribution of the maximum block-averaged concentration changes within the gray block shown in Panel B across all the channels, both for HbO2 (top) and HbR (bottom). The bilateral occipital cortices consistently respond to the full flashing checkerboard, as shown by the more red for HbO2 and more blue HbR channels. [Pinti et al., 2020]

7 Ethical Considerations

This project is a non-medical, educational study involving adult student participants from Luxembourg. It uses fNIRS solely for cognitive assessment in an academic context, and not as a medical device or diagnostic tool. The fNIRS technology is non-invasive and considered safe for participants. Overall, the study's risk profile is minimal, comparable to typical classroom activities. We will adhere to strict ethical standards as outlined below:

- **Educational Scope and Minimal Risk:** No medical procedures or treatments are involved. The fNIRS device will only monitor brain activity during problem-solving tasks, without any clinical intent. As a non-invasive technique, fNIRS poses no physical harm (it emits only low-intensity light and no radiation), and participants will engage in normal exam-like questions. The study is classified as minimal risk, with no expected adverse effects beyond potentially mild fatigue or discomfort from wearing the fNIRS headband.
- **Compliance with CEP Guidelines:** The research will comply with the guidelines of the Psychology Research Ethics Committee (*Commissie Ethiek Psychologie*, CEP, Netherlands) for psychophysiological assessments, including fNIRS. CEP is the more up-to-date guide, as it includes special considerations for fNIRS research. In line with CEP recommendations, we emphasize in the informed consent that:

  - No medical statements can be made based on the data collected;
  - Participants who wish to obtain insights into their physical functioning should consult a medical professional.

  These statements are included in the consent documents and will be clearly communicated to participants before participation.

- **Informed Consent:** All participants will be volunteers who provide written informed consent prior to participation. They will receive a clear explanation of the study's purpose, procedures, and any potential risks or inconveniences. Participants will be informed of their right to withdraw from the study at any time without penalty or loss of benefits. Consent forms will emphasize that the fNIRS device is used for research purposes only (not for medical diagnosis) and that participation is entirely voluntary. The data will be deleted after 5 years. Participants will be informed about the right to erase their data at any time.
- **GDPR Compliance and Data Pseudonymization:** The collection and handling of personal data will strictly follow EU General Data Protection Regulation (GDPR) guidelines. Only minimal personal data (e.g. gender, age, study program, parents' occupation) will be collected, and all such data will be pseudonymized, see Section 12 Demographics Questionnaire. Each participant will be assigned an ID code so that datasets contain no directly identifying information. Personal details and consent forms will be stored securely and separately from research data. All data will be stored on secure servers located within the EU, and **no** personal data will be transferred outside the EU. Access to participant data will be limited to the research team, and data will be encrypted/protected to ensure confidentiality.
- **No Audio/Video/Geolocation Recording:** The study will not involve any video recording, audio recording, or geolocation tracking of participants. The only data gathered will be: (a) fNIRS signals reflecting brain activity, and (b) participants' responses and performance on exam-style questions. Neither of these data types can identify individuals. By avoiding any form of multimedia or location recording, we protect participants' privacy and ensure anonymity during analysis and any later reporting.
- **Use of Anonymized Public Dataset:** The research will utilize existing datasets (e.g., sample questions) to complement the experimental data, Section 13 Example of Assessment Question.
- **Avoidance of High-Risk Profiling or Intrusive Methods:** The study focuses on general cognitive responses and avoids any intrusive questioning or high-risk profiling of participants. We do not collect any sensitive psychological profiles, health information, or any data unrelated to the educational research question. Participants are only asked to solve standard academic problems; there is no deception, and no covert data collection. Brain-computer interaction in this context simply involves measuring brain signals during task completion, which is non-intrusive. The analysis will be performed at the group level and will not single out individuals in any harmful or stigmatizing way.
- **No Incidental Findings or Additional Risks:** Since the fNIRS device is not used for medical examination, there are no expected incidental findings (e.g., unexpected health issues) to report. Participants will not receive any clinical feedback, as the device cannot diagnose medical

conditions. The procedures pose no significant risk to participants' health or well-being, and are comparable to everyday learning activities. There are also no foreseeable risks to third parties, as no information about anyone other than the participant is collected and the experiment's effects do not extend beyond the participant themselves.

All procedures and materials will undergo ethical review and approval by the appropriate institutional review board (e.g., the LISER Ethics Committee) before the study begins. By adhering to these safeguards, the research team ensures full compliance with institutional ethical guidelines, legal requirements, and respect for the rights and welfare of all participants.

8 Participant Recruitment Strategy

To ensure a controlled and academically relevant participant pool, we will recruit **Master's students and PhD candidates from the University of Luxembourg**. This group provides an ideal initial sample for the pilot phase due to their familiarity with exam-style questions, availability, and prior exposure to abstract and concrete reasoning tasks.

Recruitment will focus on students from the **Faculty of Humanities, Education and Social Sciences** and the **Faculty of Science, Technology and Medicine (Computer Science)**. These groups offer a balanced range of academic backgrounds relevant to the cognitive skills under investigation.

Communication Strategy: Participants will be invited through multiple outreach channels:

- Targeted emails sent via the official student mailing lists of the two faculties;
- Printed posters placed on high-traffic noticeboards across relevant departments and student spaces;
- Announcements shared with course coordinators and academic supervisors to encourage dissemination through lectures and seminars.

All participants will be screened for the following eligibility criteria:

- Enrollment in a Master's or Doctoral program;
- No known neurological or cognitive disorders;
- Willingness to participate in a neuroimaging-based educational study, confirmed through written informed consent.

Participation will be entirely voluntary and compensated with a fixed fee. Informed consent will be obtained in accordance with ethical research guidelines, which will be submitted in due time to the Ethics Committee of LISER. This recruitment strategy ensures both scientific rigor and practical feasibility for the pilot phase.

9 Budget Overview (Total: €2,750)

- **Participant Compensation (€50 x 35 participants):** €1,750
- **Materials & Equipment:** €1,000

- Acquisition of two anatomically detailed brain models featuring labeled cortical regions to support participant instruction and neuroanatomical referencing during experimental sessions.
- Procurement of a dedicated mannequin headform designed for fNIRS equipment calibration, optode placement training, and experimental protocol validation.

10 Future Research Plan

This pilot study will serve as a foundation for a larger research initiative exploring neurocognitive engagement in educational settings and AI. Findings will be leveraged to secure external grants and advance methodological approaches in experimental education research. Results will be disseminated, among others, to the representatives of the Minsitry of Education (MENJE), with whom we are currently collaborating on a number of education projects, including Excellent Innovation LISER grant for AIDCO project.

11 Conclusion

By combining neuroimaging, behavioral research, and AI-driven language analysis, this project presents a unique opportunity to enhance our understanding of cognitive processing during learning and assessment. The insights gained will inform future educational policies and assessment methodologies, improving learning outcomes for students at varying cognitive levels. Within LISER context, the findings will be of high relevance to major research initiatives, such as UPSKILL that aims at improving learning and skills acquisition, among other goals.

12 Demographics Questionnaire

Participant ID: _____ Date: _____

1. **Age:** _____ years old
2. **Gender:**
   ☐ Female ☐ Male
   ☐ Non-binary / Other ☐ Prefer not to say
3. **Current level of education:**
   ☐ Bachelor's degree ☐ Master's degree (MA/MSc)
   ☐ PhD / Doctorate ☐ Other: _____
4. **Field of study:**
   _____
5. **Mother tongue / native language:**
   _____
6. **Have you previously participated in a study using neuroimaging (e.g., fNIRS, EEG, fMRI)?**
   ☐ Yes ☐ No
7. **Mother's educational level:**
   _____
8. **Father's educational level:**
   _____

Your responses will remain confidential and will only be used in aggregated, pseudonymized form for research purposes.

13 Example of Assessment Question

Example 1: A new economy emerged on a global scale in the last quarter of the 20th century. I refer to it as informational, global, and networked in order to identify its fundamental and distinctive characteristics and to emphasize its interconnected nature. It is informational because it essentially depends on its capacity to generate, process, and efficiently apply knowledge-based information. It is global because its components are organized on a global scale, either directly or through a network of connections among economic agents. It is networked because it operates through a global network of interaction among business networks.

Which structural change is a result of the form of economic organization described in the text?

A. Mass production.

B. Expansion of inventories.

C. Weakening of cartels.

D. Standardization of goods.

E. Deterritorialization of production.

Example 2:The advent of the *polis* and the birth of philosophy: between these two phenomena, the links are too strong for rational thought not to appear, in its origins, as closely connected to the social and mental structures unique to the Greek city. Thus situated in history, philosophy sheds the character of absolute revelation that it has sometimes been given, recognizing instead that in the young science of the Ionians one can see a form of reason that took shape within time. The Milesian school did not witness the birth of Reason; it built a Reason, an initial form of rationality. This Greek reason is not the experimental reason of contemporary science.

The links between the phenomena described in the passage were strengthened by the emergence of which category of thinkers?

A. The Epicureans, involved with the ideal of a happy life.

B. The Stoics, dedicated to overcoming misfortunes.

C. The Sophists, committed to the teaching of rhetoric.

D. The Peripatetics, focused on the dynamics of teaching.

E. The rhapsode poets, responsible for the narration of myth.

Example 3: Antimicrobials are natural or synthetic substances that have the ability to kill or inhibit the growth of microorganisms. The table below presents a list of hypothetical antimicrobials, along with their actions and effects on microbial metabolism.

Antimicrobial	Action	Effect
1	Binds to ribosomes	Inhibits protein synthesis
2	Binds to microtubules	Prevents chromatid segregation
3	Binds to plasma membrane phospholipids	Reduces plasma membrane permeability
4	Interferes with thymine synthesis	Inhibits DNA synthesis
5	Interferes with uracil synthesis	Prevents RNA synthesis

Which antimicrobial should be used to treat a fungal infection without affecting the normal bacterial microbiota of the body?

A. 1

B. 2

C. 3

D. 4

E. 5

Example 4: A building with 9 floors and 8 two-bedroom apartments per floor has all its apartments available for sale. The apartments are identified by two-digit numbers: the tens digit indicates the floor, and the units digit (from 1 to 8) distinguishes the apartments on the same floor. Regarding sunlight exposure, the following characteristics are observed according to the unit number:

- Apartments ending in 1 or 2: both bedrooms receive sunlight only in the morning.
- Apartments ending in 3, 4, 5, or 6: only one bedroom receives sunlight in the morning.
- Apartments ending in 7 or 8: both bedrooms receive sunlight only in the afternoon.

A person wants to buy **2 apartments on the same floor** but wants both to have at least one bedroom that gets sunlight in the morning.

How many different ways can this person choose 2 apartments that meet this condition?

Options:

A. $9 \times \dfrac{6!}{(6-2)!}$

B. $9 \times \dfrac{6!}{(6-2)! \times 2!}$

C. $9 \times \dfrac{4!}{(4-2)! \times 2!}$

D. $9 \times \dfrac{2!}{(2-2)! \times 2!}$

E. $9 \times \left(\dfrac{8!}{(8-2)! \times 2!} - 1 \right)$

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